Monte Carlo algorithm

A **Monte Carlo algorithm** is an **algorithm** for computers. It is used to simulate the behavior of other systems. It is not an exact method, but a heuristically one. Usually it uses randomness and statistics to get a result. It is a computation process that uses random numbers to produce an outcome(s).

Monte Carlo methods use repeated random sampling to solve complex problems involving uncertainty, probability, and optimization in artificial intelligence. Monte Carlo Simulation is a method used to predict and understand the behaviour of systems involving uncertainty.

Monte Carlo methods solve difficult AI problems by repeatedly using random simulations to estimate the probability or expected quality of different outcomes.

How Does Monte Carlo Simulation Work?

- Defining the Problem: Identify the process we want to simulate. Ex:- stock prices, engineering models
- Model the Inputs: If we're modeling stock prices, we might use a normal distribution to represent price fluctuations.
- Run Simulations: Generate random inputs based on the chosen probability distributions and repeat the process many times.
- Analyze the Results: After simulations, we'll get a distribution of outcomes. Possible range of scenarios not just a single expected outcome.

Given a function f(x) and a domain D, the expected value E[f(x)] can be estimated using the following formula:

$$E[f(x)] \approx \frac{1}{N} \sum_{i=1}^{N} f(x_i)$$

where:

- N is the number of random samples.
- x_i are the random samples drawn from the domain D.

Applications in AI

- **Game playing:** Chess, Go, Poker, etc.
- **Reinforcement learning:** Estimating returns and action values.
- **Probabilistic inference:** Estimating probabilities in complex models.
- **Robotics:** Planning under uncertainty.
- **Optimization:** Finding good solutions in large search spaces.
- **Decision-making:** Evaluating uncertain outcomes.

Problem: Ramesh is a dentist who scheduled all his patients for 30 min appointments. Some patients take more or less than 30 min depending on the type of dental work to be done. The following summary shows various categories of work, their probability, and the time actually needed to complete work. Simulate the dentist's clinic for 4 hrs and find out the average waiting time for the patient as well as the ideal time of doctor. Assume that all patient reaches at clinic exactly as per schedule.

Category	Time Required (min)	No. of Patients	Probability
Filling	45	30	0.30
Crowning	60	15	0.15
Cleaning	15	10	0.10
Extracting	45	25	0.25
Checkup	15	20	0.20
Total	—	**100**	**1.00**

Using Random numbers: - 42, 82, 10, 34, 25, 66, 17, 79

"No. of patient visiting within 4hrs are 8 patient."

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Category	Probability	Cumm. Probability	Random No. of Interval
Filling	0.3	0.3	0 - 29
Crowning	0.15	0.45	30 - 44
Cleaning	0.10	0.55	45 - 54
Extracting	0.25	0.8	55 - 79
Checkup	0.2	1.0	80 - 99

Patient	Schedule Arrival	Random No.	Category	Service Time Needed
1	8:00	42	Crowning	60
2	8:30	82	Check up	15
3	9:00	10	Filling	45
4	9:30	34	Crowning	60
5	10:00	25	Filling	45
6	10:30	66	Extracting	45
7	11:00	17	Filling	45
8	11:30	79	Extracting	45

Patient	Arrival	Service Start	Service Duration	Service End	Waiting Time (min)	Idle time For doctor
1	8:00	8:00	60	9:00	0	0
2	8:30	9:00	15	9:15	30	0
3	9:00	9:15	45	10:00	15	0
4	9:30	10:00	60	11:00	30	0
5	10:00	11:00	45	11:45	60	0
6	10:30	11:45	45	12:30	75	0
7	11:00	12:30	45	1:15	90	0
8	11:30	1:15	45	2:00	105	0

$$\Sigma t = 405 \text{ min}$$

$$\text{avg} = \frac{\Sigma t}{n} = \frac{405}{8} = 50.625 \text{ min}$$